2021 Ph H2 Q3

Section: Our Dynamic Universe

Topic: Collisions, Explosions & Impulse

A steel ball (mass 1.59×10^{-2} kg) is dropped from rest from 1.27 m onto a Perspex lens. (a) Find the impact speed. (b) Given $|\Delta p| = 0.14$ kg m s^{-1}, find the rebound speed. (c) Define an inelastic collision. (d) Explain why a softer lens is less likely to break.

Worked solution

(a) Impact speed

From energy or $v^2 = u^2 + 2gh$ with $u = 0$:
$v = \sqrt{(2gh)} = \sqrt{(2\times9.8\times1.27)} = 4.99$ m s^{-1}.

Answer: 4.99 m s^{-1}

(b) Rebound speed

Take downward as positive just before impact, upward as positive just after.
Magnitude of momentum change: $|\Delta p| = m(v_{down} + v_{up}) \Rightarrow v_{up} = |\Delta p|/m - v_{down}$.
$v_{up} = 0.14 / (1.59\times10^{-2}) - 4.99 = 3.82$ m s^{-1} (upwards).

Answer: 3.82 m s^{-1} (upwards)

(c) Inelastic collision

A collision in which total kinetic energy is not conserved (some is transformed to heat, sound, or deformation). Momentum of the system is still conserved.

(d) Softer lens less likely to break

A softer material deforms more during impact, increasing the contact time. For a given impulse (Δp), the average force $F = \Delta p/\Delta t$ is reduced. The material also absorbs more energy through plastic deformation, lowering peak stress and reducing the chance of fracture.

Final answers

(a) $v \approx 4.99\ \mathrm{m\ s^{-1}}$

(b) Rebound speed ≈ $3.82\ \mathrm{m\ s^{-1}}$ (upwards)

(c) Inelastic: KE not conserved (momentum conserved)

(d) Softer lens increases Δt and absorbs energy ⇒ smaller peak force ⇒ less likely to break

Revision tips

- Free fall from rest: $v = \sqrt{2gh}$.
- Impulse equals change in momentum; reversal often gives $\Delta p = m(v_{down} + v_{up})$.

- Momentum is always conserved in collisions; KE only in elastic ones.
- Increasing stopping time reduces average force for a given impulse.